

Math 126 Exam 2 Rubric

Math 126 - Spring 2026
Exam 2
May 21, 2026

Name RUBRIC

Student ID # _____

Section _____

HONOR STATEMENT

"I affirm that my work upholds the highest standards of honesty and academic integrity at the University of Washington, and that I have neither given nor received any unauthorized assistance on this exam."

SIGNATURE: _____

- This exam consists of this cover, four pages of questions, and a blank "scratch sheet". If you put work on the scratch sheet and you want it to be graded, then you must clearly tell us in the problem to "see scratch page".
- You will have 50 minutes.
- You are allowed to use a Ti-30x IIS Calculator model ONLY (**no other calculators allowed**) and one 8.5 by 11 inch sheet of handwritten notes (front and back). All other sources are forbidden.
- Turn your cell phone OFF and put it away for the duration of the exam. You may not listen to headphones or earbuds during the exam.
- **You must show your work.** The correct answer with no supporting work may result in no credit.
- Leave your answer in exact form. Simplify standard trig, inverse trig, natural logarithm, and root values. Here are several examples: you should write $\sqrt{4} = 2$ and $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$ and $\ln(1) = 0$ and $\tan^{-1}(1) = \frac{\pi}{4}$.
- Unless otherwise indicated, when rounding is necessary, you may round your final answer to two digits after the decimal.
- **Do not write within 1 centimeter of the edge!** Your exam will be scanned for grading.
- There may be multiple versions, you have signed an honor statement, and cheating is a hassle for everyone involved. JUST DO NOT CHEAT.

GOOD LUCK!

1. (12 pts)

- 6 (a) A drone is flying in the air following the path $\mathbf{r}(t) = \langle 4 \cos(t), 4 \sin(t), 4 + t^3 - 12t \rangle$. Find all values of t at which the tangential component of acceleration, a_T , is equal to zero.

$$a_T = \frac{\dot{\mathbf{r}}'(t) \cdot \ddot{\mathbf{r}}''(t)}{|\dot{\mathbf{r}}'(t)|} \stackrel{?}{=} 0 \Leftrightarrow \dot{\mathbf{r}}'(t) \cdot \ddot{\mathbf{r}}''(t) = 0$$

$$\dot{\mathbf{r}}'(t) = \langle -4 \sin(t), 4 \cos(t), 3t^2 - 12 \rangle$$

$$\ddot{\mathbf{r}}''(t) = \langle -4 \cos(t), -4 \sin(t), 6t \rangle$$

$$\cancel{16 \cos(t) \sin(t)} - \cancel{16 \cos(t) \sin(t)} + 6t(3t^2 - 12) \stackrel{?}{=} 0$$

$$18t(t^2 - 4) \stackrel{?}{=} 0 \begin{matrix} \nearrow t=0 \\ \rightarrow t=-2 \\ \searrow t=2 \end{matrix}$$

$$-2, 0, 2$$

List all answers $t =$ _____

- 6 (b) Let $z = f(x, y)$ be defined implicitly by $xz + z^2 + y^2 = 6$. Find the equation for the tangent plane at $(x, y, z) = (1, 2, 1)$.

$$\frac{\partial z}{\partial x} : x \cdot \frac{\partial z}{\partial x} + z + 2z \frac{\partial z}{\partial x} + 0 = 0$$

$$@ (x, y, z) = (1, 2, 1) \Rightarrow 1 \frac{\partial z}{\partial x} + 1 + 2 \frac{\partial z}{\partial x} = 0 \Rightarrow 3 \frac{\partial z}{\partial x} = -1 \Rightarrow \frac{\partial z}{\partial x} = -\frac{1}{3}$$

$$\frac{\partial z}{\partial y} : x \frac{\partial z}{\partial y} + 2z \frac{\partial z}{\partial y} + 2y = 0$$

$$@ (x, y, z) = (1, 2, 1) \Rightarrow 1 \frac{\partial z}{\partial y} + 2 \frac{\partial z}{\partial y} + 4 = 0 \Rightarrow 3 \frac{\partial z}{\partial y} = -4 \Rightarrow \frac{\partial z}{\partial y} = -\frac{4}{3}$$

$$z - 1 = -\frac{1}{3}(x - 1) - \frac{4}{3}(y - 2)$$

Tangent plane: _____

2. (12 pts) The two parts below are not related.

7

- (a) Find and classify all critical points for $g(x, y) = x^3 - x^2y + y^2 - 2y$. (Clearly show work using the 2nd derivative test)

$$\textcircled{1} g_x = 3x^2 - 2xy \stackrel{?}{=} 0$$

$$\textcircled{2} g_y = -x^2 + 2y - 2 \stackrel{?}{=} 0$$

$$\textcircled{1} x(3x - 2y) = 0 \begin{cases} \rightarrow x = 0 \xrightarrow{\textcircled{2}} 2y - 2 = 0 \rightarrow y = 1 \quad (0, 1) \\ \rightarrow 3x - 2y = 0 \xrightarrow{\textcircled{2}} -x^2 + 3x - 2 = 0 \end{cases}$$

$$\begin{aligned} y &= \frac{3}{2}x \\ 0 &= x^2 - 3x + 2 \\ 0 &= (x-1)(x-2) \end{aligned} \begin{cases} \nearrow x=1 \quad (1, \frac{3}{2}) \\ \searrow x=2 \quad (2, 3) \end{cases}$$

$$\begin{aligned} g_{xx} &= 6x - 2y \\ g_{yy} &= 2 \\ g_{xy} &= -2x \end{aligned}$$

$$(0, 1) \rightarrow g_{xx} = -2, g_{yy} = 2, g_{xy} = 0, D = -4 < 0$$

$$(1, \frac{3}{2}) \rightarrow g_{xx} = 3, g_{yy} = 2, g_{xy} = -1, D = 5 > 0 \quad \begin{matrix} g_{xx} > 0 \\ g_{yy} > 0 \end{matrix}$$

$$(2, 3) \rightarrow g_{xx} = 6, g_{yy} = 2, g_{xy} = -4, D = -4 < 0$$

LOCAL MIN
 $(1, \frac{3}{2})$

SADDLE POINTS
 $(0, 1) \quad (2, 3)$

Labeled List of Critical Points:

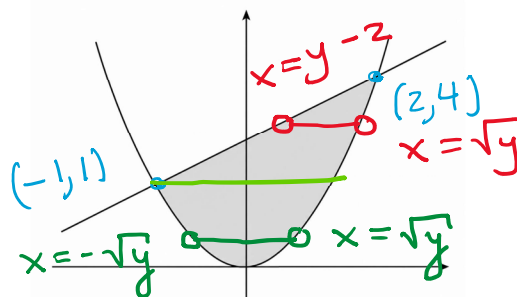
5

- (b) Let D be the region shown below, bounded by $y = x^2$ and $y = x + 2$.

Set up $\iint_D f(x, y) dA$ in the order $dx dy$. Do NOT evaluate.

(Hint: This might require splitting it into two integrals!)

$$\begin{aligned} x^2 &\stackrel{?}{=} x + 2 \\ x^2 - x - 2 &= 0 \\ (x-2)(x+1) &= 0 \end{aligned}$$



$$dx dy: \int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} f(x, y) dx dy + \int_1^4 \int_{y-2}^{\sqrt{y}} f(x, y) dx dy$$

3. (14 pts) The two parts below are not related.

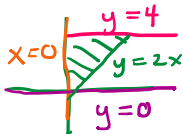
6

- (a) Find the volume of the solid that is bounded above by $z = y - 2x$, bounded below by $z = 0$ and between the planes $y = 4$, $x = 0$, and $y = 0$. Set up and evaluate a double integral.

I $\iint_R y - 2x \, dA$

II

III $\int_0^4 \int_{2x}^4 y - 2x \, dy \, dx$ or $\int_0^4 \int_0^{y/2} y - 2x \, dx \, dy$



← GIVEN AND NEED WHEN $z = y - 2x$
AND $z = 0$
INTERSECT!
SO $0 = y - 2x$
 $y = 2x$

$$\begin{aligned} &= \int_0^4 \int_0^{y/2} y - 2x \, dx \, dy \\ &= \int_0^4 \left[yx - x^2 \right]_0^{y/2} dy \\ &= \int_0^4 \left[\frac{1}{2}y^2 - \frac{1}{4}y^2 \right] dy \\ &= \int_0^4 \frac{1}{4}y^2 dy = \frac{1}{12}y^3 \Big|_0^4 = \frac{4^3}{12} = \frac{16}{3} \end{aligned}$$

Volume = $\frac{16}{3} = 5.3\bar{3}$

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- (b) Let R be the region in the first quadrant of the xy -plane that is inside the circle $x^2 + y^2 = 9$ and outside the cardioid $r = 2 + 2\cos(\theta)$ (shaded below).

Using polar coordinates, evaluate $\iint_R \frac{y}{x^2 + y^2} \, dA$.

$x^2 + y^2 = 9 \Rightarrow r = 3$

INTERSECTION

$3 = 2 + 2\cos(\theta) \Rightarrow 1 = 2\cos(\theta) \Rightarrow \frac{1}{2} = \cos(\theta)$
 $\theta = \pi/3$

$\int_{\pi/3}^{\pi/2} \int_{2+2\cos\theta}^3 \frac{\sin\theta}{r^2} \cdot r \, dr \, d\theta$

$= \int_{\pi/3}^{\pi/2} \sin\theta \cdot r \Big|_{2+2\cos\theta}^3 d\theta$

$= \int_{\pi/3}^{\pi/2} 3\sin\theta - \sin\theta (2 + 2\cos\theta) d\theta$

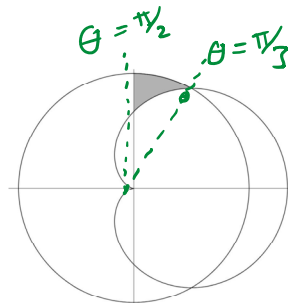
$= \int_{\pi/3}^{\pi/2} \sin\theta - 2\cos\theta \sin\theta d\theta$

$u = \sin\theta$
 $du = \cos\theta d\theta$

$\begin{cases} \int 2\cos\theta \sin\theta d\theta \\ = \int 2u du \\ = u^2 + C \\ = \sin^2\theta + C \end{cases}$

$= -\cos\theta - \sin^2\theta \Big|_{\pi/3}^{\pi/2}$

$= (0 - 2) - \left(-\frac{1}{2} - \left(\frac{\sqrt{3}}{2}\right)^2\right) = -2 + \frac{1}{2} + \frac{3}{4} = -2 + \frac{5}{4} = \frac{1}{4}$



$\iint_R \frac{y}{x^2 + y^2} \, dA = \frac{1}{4}$

4. (12 pts) You are designing the wooden open-top Bento box shown below. The box has length x , width y , height z , one divider parallel to the side of length y , and one divider of length $x/2$.

The total volume must be 60 cubic inches. Find the dimensions that minimize the total amount of wood used.

(Hint: The wood used consists of the bottom, four outside walls, and two divider walls. You must set up a two-variable function for the total surface area of the wood and find its critical point, but you do NOT have to use the second derivative test.)

$$xyz = 60$$

$$\Rightarrow z = \frac{60}{xy}$$

$$S.A. = xy + 3yz + 2xz + \frac{x}{2}z$$

$$= xy + 3yz + 2.5xz$$

6 $f(x,y) = xy + \frac{180}{x} + \frac{150}{y}$

$$\textcircled{1} f_x = y - \frac{180}{x^2} \stackrel{?}{=} 0 \Rightarrow y = \frac{180}{x^2}$$

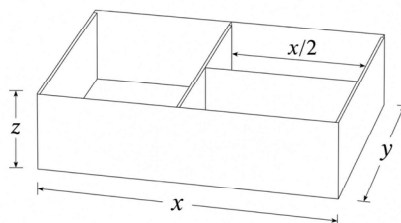
$$\textcircled{2} f_y = x - \frac{150}{y^2} \stackrel{?}{=} 0 \Rightarrow x = \frac{150}{y^2} \Rightarrow xy^2 = 150$$

$$\textcircled{1} y = \frac{180}{x^2} \xrightarrow{\textcircled{2}} x \left(\frac{180}{x^2} \right)^2 = 150$$

$$\cancel{x} \frac{180^2}{x^4} = 150$$

$$\Rightarrow 180^2 = 150x^3$$

$$\Rightarrow x^3 = \frac{180^2}{150} = 216 \Rightarrow x = 6$$



6

NOT REQUIRED

$$f_{xx} = \frac{360}{x^3}, f_{yy} = \frac{300}{y^3}, f_{xy} = 1$$

$$D = \frac{360}{6^3} \cdot \frac{300}{5^3} - 2^2 = 3 > 0 \quad \begin{matrix} f_{xx} > 0 \\ f_{yy} > 0 \end{matrix} \text{ LOCAL MIN!}$$

$$y = \frac{180}{6^2} = 5$$

$$z = \frac{60}{xy} = 2$$

$$(6, 5, 2)$$

Dimensions: $(x, y, z) =$ _____

You may use this page for scratch-work or extra room.

All work on this page will be ignored unless you write and circle “see scratch page” on the problem and you label your work.